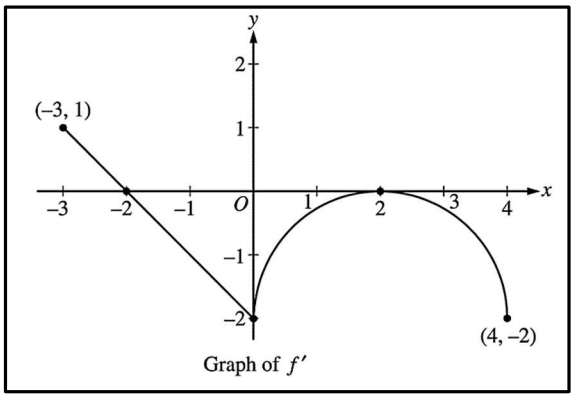
Worksheet 6-4 & 6-5 C
FTC Part 2 and Graphical Analysis #3
AP Calculus

Name SOLUTIONS

1.) **2003 AB 4**

Let f be a function defined on the closed interval $-3 \le x \le 4$ with $f(0) = 3$. The graph of f', the derivative of f, consists of one line segment and a semicircle, as shown at the right.



Graph of f'

(a) On what intervals, if any, is f increasing? Justify your answer.

f is increasing on $[-3, -2]$ because $f'(x) \ge 0$ on this interval.

(b) Find the x coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.

f has a point of inflection at $x = 0$ because $f'(x)$ changes from decreasing to increasing at this x-value, and at $x = 2$ because $f'(x)$ changes from increasing to decreasing at this x-value.

(c) Find an equation for the line tangent to the graph of f at the point $(0, 3)$.

$f'(0) = -2$ so EQ_{tan}: $y - 3 = -2(x - 0)$

(d) Find $f(-3)$ and $f(4)$. Show the work that leads to your answers.

$f(0) = 3$

$\int_{-3}^{0} f'(x)\,dx = f(0) - f(-3)$

$f(-3) = f(0) - \int_{-3}^{0} f'(x)\,dx = 3 - \left[\frac{1}{2}(1)(1) - \frac{1}{2}(2)(2)\right]$

$= 3 - \frac{1}{2} + 2 = 4.5$

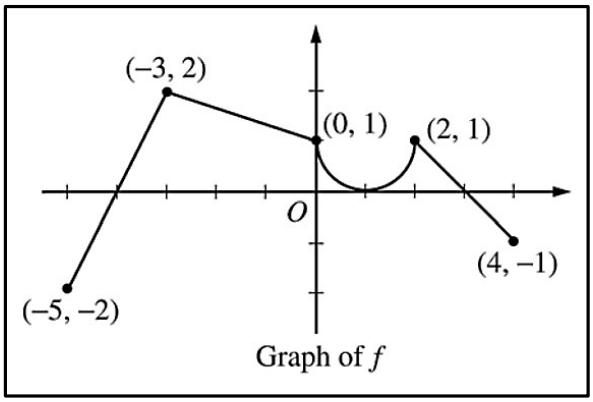
$\int_{0}^{4} f'(x)\,dx = f(4) - f(0)$

$f(4) = f(0) + \int_{0}^{4} f'(x)\,dx = 3 + \left[-\left(4(2) - \frac{1}{2}\pi(2)^2\right)\right]$

$= 3 - 8 + 2\pi = -5 + 2\pi$

2.) **2004 AB5**

The graph of the function f shown at the right consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^{x} f(t)dt$



Graph of f

(a) Find $g(0)$ and $g'(0)$

$$g(0) = \int_{-3}^{0} f(t)dt = 3(1) + \tfrac{1}{2}(3)(1)$$
$$= 3 + \tfrac{3}{2} = \tfrac{9}{2}$$

$$g'(0) = f(0) = 1$$

(b) Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Justify your answer.

g attains a relative maximum at $x = 3$ because $g'(x) = f(x)$ changes from positive to negative at this x-value.

(c) Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.

The absolute minimum value of g on $[-5, 4]$ occurs at either endpoint or the critical numbers of $g(x)$ in the interval where $g'(x) = f(x) = 0$ and $g'(x) = f(x)$ changes from negative to positive, which is at $x = -4$.

x	$g(x)$
-5	0
-4	-1
4	$\frac{13}{2} - \frac{\pi}{2}$

$$g(-5) = \int_{-3}^{-5} f(t)dt = -\int_{-5}^{-3} f(t)dt$$
$$= -\left[\tfrac{-1}{2}(1)(2) + \tfrac{1}{2}(1)(2)\right] = -\left[-1+1\right] = 0$$

$$g(-4) = \int_{-3}^{-4} f(t)dt = -\int_{-4}^{-3} f(t)dt$$
$$= -\left[\tfrac{1}{2}(1)(2)\right] = -(1) = -1$$

$$g(4) = \int_{-3}^{4} f(t)dt = 3 + \tfrac{3}{2} + 2 - \tfrac{1}{2}\pi(1)^2 + \tfrac{1}{2} - \tfrac{1}{2}$$
$$= \tfrac{13}{2} - \tfrac{\pi}{2}$$

The absolute minimum value of g on $[-5, 4]$ is -1 at $x = -4$

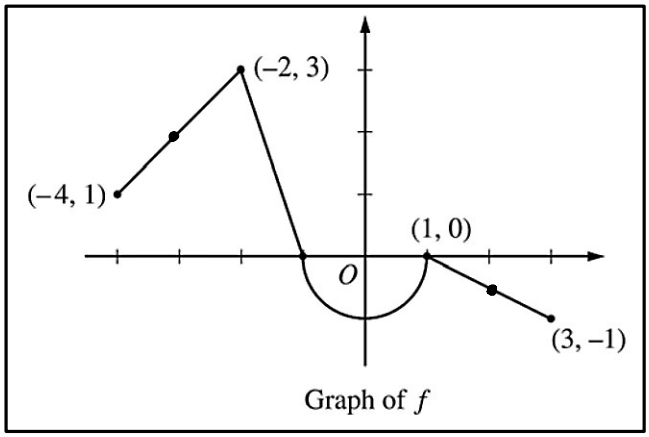
(d) Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

g has a point of inflection at $x = -3, x = 1$, and $x = 2$ ($g'(x) = f(x)$ changes from increasing to decreasing or decreasing to increasing at these x-values).

3.) **2012 AB3**

Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given at the right.

Let g be the function given by $g(x) = \int_1^x f(t)dt$



Graph of f

(a) Find the values of $g(2)$ and $g(-2)$.

$$g(2) = \int_1^2 f(t)\,dt = -\frac{1}{2}(1)\left(\frac{1}{2}\right)$$
$$= -\frac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t)\,dt = -\int_{-2}^{1} f(t)\,dt = -\left[\frac{1}{2}(1)(3) - \frac{1}{2}\pi(1)^2\right] = -\left[\frac{3}{2} - \frac{\pi}{2}\right]$$
$$= \frac{-3}{2} + \frac{\pi}{2}$$

(b) For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

$$g'(-3) = f(-3) = 2$$
$$g''(-3) = f'(-3) = \frac{3-1}{-2--4} = \frac{2}{2} = 1$$

(c) Find the x coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor maximum at the point. Justify your answers.

g has a horizontal tangent line when $g'(x) = f(x) = 0$
this occurs at $x = -1$ and $x = 1$

g has a relative maximum at $x = -1$ because $g'(x) = f(x)$ changes from positive to negative at this x-value.

g has neither a relative min nor max at $x = 1$ because $g'(x) = f(x)$ does not change signs at this x-value.

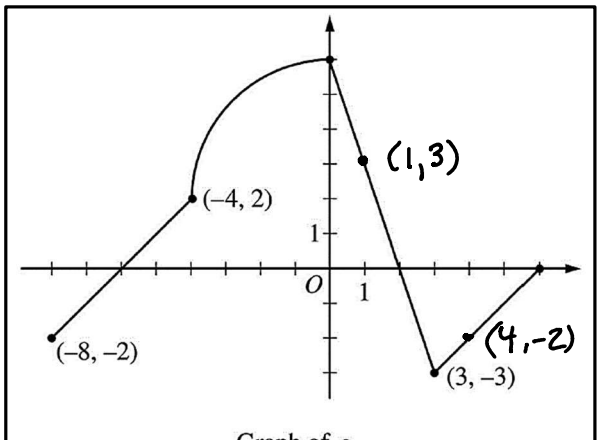
(d) For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

g has a point of inflection at $x = -2$ and $x = 1$ because $g'(x) = f(x)$ changes from increasing to decreasing at these x-values, and at $x = 0$ because $g'(x) = f(x)$ changes from decreasing to increasing at this x-value.

4.) **2016 PE AB3**

A continuous function g is defined on the closed interval $-8 \le x \le 6$. The graph of g, shown above, consists of three line segments and a quarter circle centered at the point $(0, 2)$. Let f be the function given by

$$f(x) = \int_{-8}^{x} g(t)dt.$$



Graph of g

(a) Find all values of x in the interval $-8 < x < 6$ at which f has a critical point. Classify each critical point as the location of a local minimum, a local maximum, or neither. Justify your answers.

f has critical points at $x = -6$ and $x = 2$

$(f'(x) = g(x) = 0)$

At $x = -6$, f has a relative minimum because $f'(x) = g(x)$ changes from negative to positive.

At $x = 2$, f has a relative maximum because $f'(x) = g(x)$ changes from positive to negative.

(b) Find $f(0)$

$$f(0) = \int_{-8}^{0} g(t)dt = -\frac{1}{2}(2)(2) + \frac{1}{2}(2)(2) + 4(2) + \frac{1}{4}(\pi)(4)^2$$

$$= -2 + 2 + 8 + 4\pi = \boxed{8 + 4\pi}$$

(c) Find $\lim_{x \to -4} \frac{f(x)}{x^2 + 4x}$

$$\lim_{x \to -4} f(x) = f(-4) = \int_{-8}^{-4} g(t)dt = -2 + 2 = 0$$

$$\lim_{x \to -4} (x^2 + 4x) = 16 - 16 = 0$$

$f'(4) = g(4)$
$= -2$

By L'Hospital's Rule:

$$\lim_{x \to -4} \frac{f(x)}{x^2 + 4x} = \lim_{x \to -4} \frac{f'(x)}{2x + 4} = \frac{f'(-4)}{-8 + 4} = \frac{2}{-4} = -\frac{1}{2}$$

(d) Let h be the function defined by $h(x) = \frac{g(x)}{x^2 + 1}$. Find $h'(1)$.

$g(1) = 3$
$g'(1) = -3$

$$h'(x) = \frac{(x^2 + 1)g'(x) - g(x) \cdot 2x}{[x^2 + 1]^2}$$

$$h'(1) = \frac{(1 + 1)g'(1) - g(1) \cdot 2}{[1 + 1]^2}$$

$$= \frac{2(-3) - (3)(2)}{2^2} = \frac{-6 - 6}{4} = \frac{-12}{4} = \boxed{-3}$$